TPSM82901SISR possible alt LMZ21701 expensive but small
TPSM336{20,10,06) slightly larger but more available, TPSM33620S3QRDNRQ1 https://jlcpcb.com/parts
TPS63070 buck/boost highly available but large

L2
1-1.5uH 1A 0603
U5
TPS63070RNMR
1-3s
3.0-12.6v (max 2.0-16v)
+5V F1 500ma
D1 13v
C23 22u 25v 0603
C5 22u 25v 0603
5V_FUSED
R6 10k
JP3 NP
JP2 JP
PS Low = Forced PWM mode
lower efficiency and ripple
C24 100n
+3.3V
3.3v 0.5A
R49 470k
R5 150k
C27 10u
C25 >47u 0603
0.8v
JP1 JP RESET
Note: !PG reset is required due to esp32
browout detection observed unreliable by itself
D2 3.3v
R14 470
D3 Power
Goal: Linear 0-27v (6s)
Safe -27-27v (TODO find max pin current)
nonlinear to ~-5v-35v (one cell off, ground lift, etc)
ADC_ATTEN_DB_0 100 mV ~ 950 mV (common, default prior to supporting vbat_atten)
ADC_ATTEN_DB_2_5 100 mV ~ 1250 mV
ADC_ATTEN_DB_6 150 mV ~ 1750 mV
ADC_ATTEN_DB_11 150 mV ~ 3100 mV (common) nonlinear after ~2.6v
47/(47+270)*3.3*1000
offset!=489.2744478000
offset=450.850
vscale=784.9479
atten 11db
~3.3hz
NOTE adc polling rate is 10hz
VBAT_SAMPLE_INTERVAL 100ms
Y1 52Mhz TCXO
C11 100n
C9 1n
RADIO_XIN
PICO-SIP pin should
already be 50R or close
enough for wifi
J9 Little Tower
Tower is for experiments with espnow.
otherwise use a trace squiggle
R48 SERIAL_TX
R47 SERIAL_RX
CF/Prog J10
R1 10k
C1 1u
C3 100n
C4 10u
RESET
R13 10k
SW2 SW_SPST
NOTE GPIO34 and up are input only
Strap MTDO 1=Debug 0=Silent
Strap MTDI 0 Flash 3.3V PD
Strap ? (SDIO) PU
Strap 1 PU
Strap 0 When Boot=0 PD
U1 ESP32-PICO-D4 or ESP32-PICO-V3 (v3 as configured)
ESP32-PICO-V3
Note: 3.3v is out of spec for led, but works.
(16v from 5v input would definitely not work)
D4 WS2812B 2020
LED_RGB
C10 C12 C7 C8
C2 >470n
L1 15u
U2 SX1281
U4 2450FM07D0034T
J13 U.FL
Layout per filter DS
Per anyleaf: When using a TCXO, leave
pin 5, normally ground, NC
NOTE: ADC2 is problematic when wifi is enabled.
since wifi is usually disabled, maybe disable adc2 functions in programming mode
or just avoid adc2
S6 J12 Balance S0
LVS1 R31 270k
LVS2 R32 270k
LVS3 R33 270k
LVS4 R34 270k
LVS5 R35 270k
LVS6 R36 270k
LVS7 R37 270k
VS_5V R38 270k
LVS1 R39 47k
LVS2 R40 47k
LVS3 R41 47k
LVS4 R42 47k
LVS5 R43 47k
LVS6 R44 47k
LVS7 R45 47k
VS_5V R46 47k
LVS1 C14 100n
LVS2 C15 100n
LVS3 C16 100n
LVS4 C17 100n
LVS5 C18 100n
LVS6 C19 100n
LVS7 C20 100n
VS_5V C21 100n
Pin Holes for Assembly Alignment
H1 MountingHole
H2 MountingHole
H3 MountingHole
C6 C13 C22 C28 C29 C30 C31
>16v
Optional capacitor positions for handling poor BEC
and servo current spikes
R15 R16 R17 R18 R19 R20 R21 R22
100
J1 J2 J3 J4 J5 J6 J7 J8
PWM

Pin	D4	V3	O2
25	Flash	NC	NC
27	Flash	IO20	IO20
28	IO9	IO9	PSRAM
29	IO10	IO10	PSRAM
30	Flash	Flash	Flash
31	Flash	-	-
32	Flash	IO7	IO7
33	Flash	IO8	IO8
35	IO18	NC	NC
36	IO23	NC	NC

Strap:

MTDI	Need 0
IO0	(Boot)
IO2	Need 0 (During Programming)
MTDO	High for rom debug
IO5	?SDIO

V2: invert serial connector pinout (to match radiomaster),
remove test points, re-layout for single sided and larger (cheaper) clearances
(v2 never built due to part availability)
V2.1: change to buck-boost regulator (more available), remove EN button to save space

Kevin Malec
Sheet: /
File: elrs-lvs.kicad_sch
Title: ELRS-LVS
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